# Kanban.PDF

Mudit Chaturvedi_PES1UG24CS279-BPS#33

SRN: PES1UG24CS279 | Name: Mudit Chaturvedi

Problem Statement #33 — Micro-Lending & Peer Credit Risk Assessor

Jira site: mudit300477.atlassian.net

Project: **Kanban_BPS#33** | Key: **KBPS33** (company-managed Kanban)

Template: Software Development — Kanban

## 1. Project setup

A company-managed Kanban project was created for Lab 1 scenario #33. The project name is Kanban_BPS#33 and the project key is KBPS33 (Jira keys cannot contain '#'). Work is visualised on the KBPS33 board with columns Backlog → Selected for Development → In Progress → Done.

## 2. Mapping Lab 1 requirements to the board

All 5 functional requirements and 2 non-functional requirements from Lab 1 were added as board work items. Each requirement is represented as one Epic (scenario), one User Story (use case), one Task, and one Sub-task.

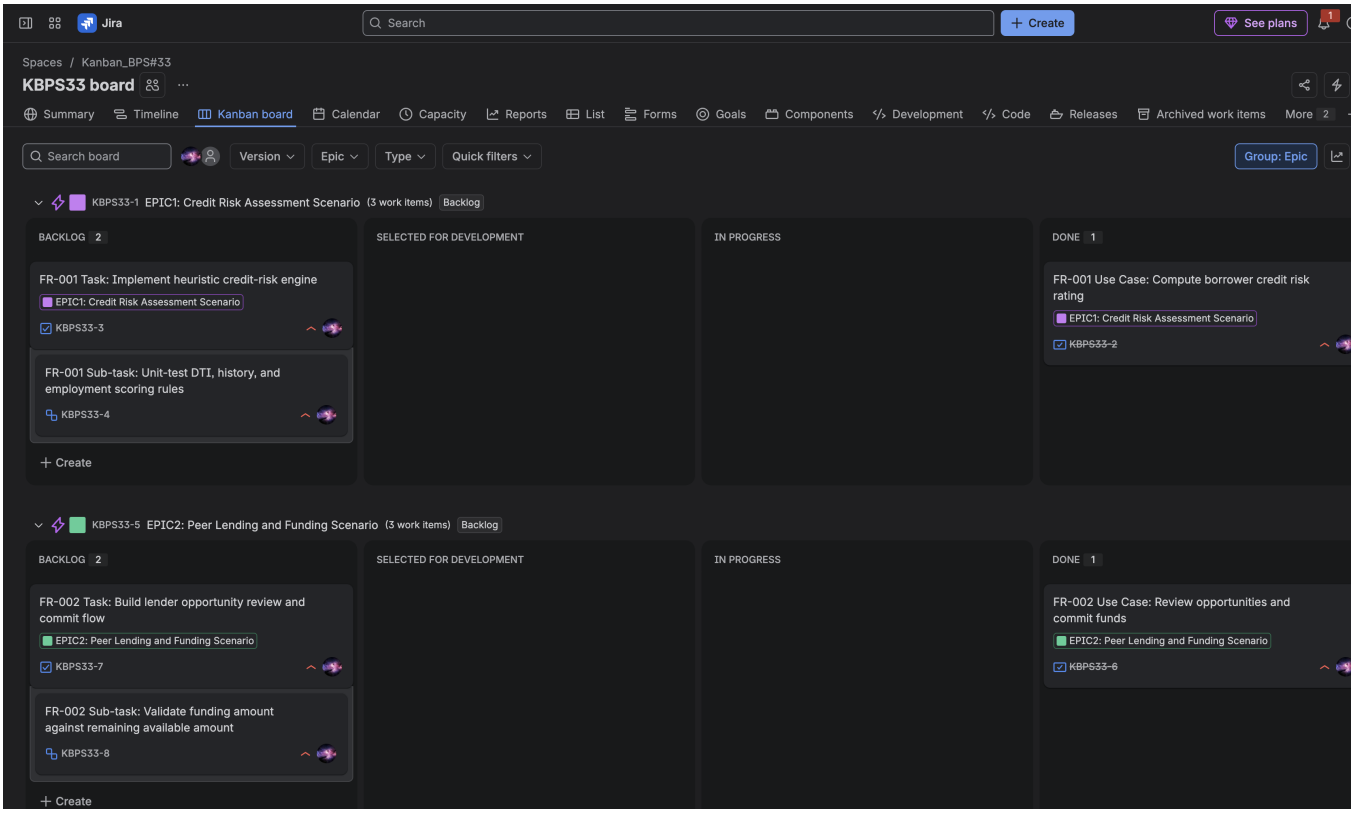



**Figure.** KBPS33 Kanban board grouped by Epic, showing FR-001 and FR-002 stories, tasks and sub-tasks moving across columns.

## 3. Epics (scenarios) assigned to board work

Seven epics capture the Micro-Lending & Peer Credit Risk Assessor scenarios:

- EPIC1 KBPS33-1 — Credit Risk Assessment Scenario (FR-001)
- EPIC2 KBPS33-5 — Peer Lending and Funding Scenario (FR-002)
- EPIC3 KBPS33-9 — Repayment Schedule Scenario (FR-003)
- EPIC4 KBPS33-13 — Loan Repayment Recording Scenario (FR-004)
- EPIC5 KBPS33-17 — Default Warning Scenario (FR-005)
- EPIC6 KBPS33-21 — Transaction Integrity Scenario (NFR-001)
- EPIC7 KBPS33-25 — Data Protection Scenario (NFR-002)

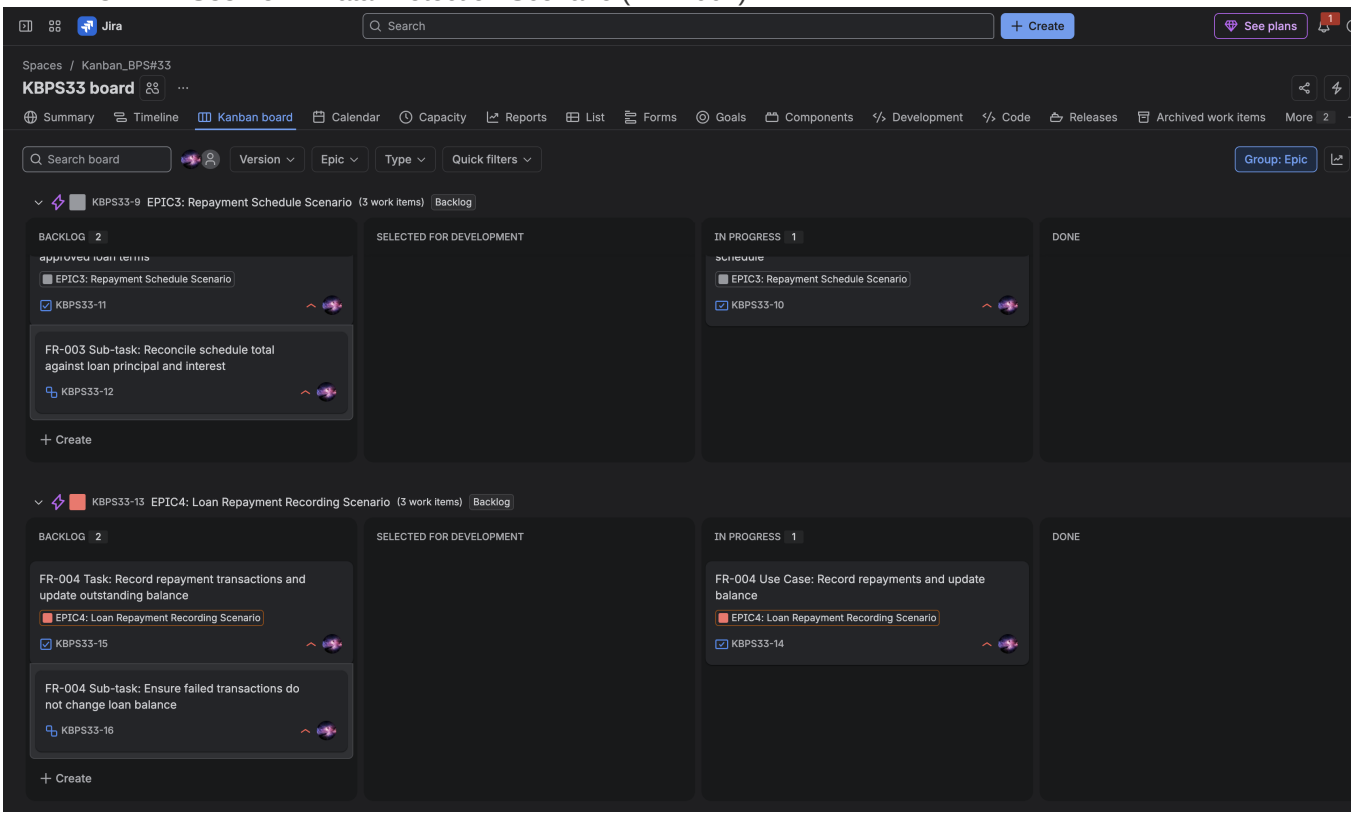



**Figure.** List view of Kanban_BPS#33: all seven epics with child use-case stories and implementation tasks, assigned to Mudit Chaturvedi.

## 4. User stories for all 7 requirements (use cases)

- FR-001 Story KBPS33-2 (8 pts): As a peer lender, I want a Low/Medium/High risk tier from DTI, repayment history and employment stability, so that I fund only eligible borrowers.
- FR-002 Story KBPS33-6 (5 pts): As a peer lender, I want to review eligible loans and commit a valid amount, so that I can participate in funding.
- FR-003 Story KBPS33-10 (5 pts): As a borrower, I want a structured installment schedule, so that I know amounts, due dates and remaining balance.
- FR-004 Story KBPS33-14 (5 pts): As a borrower, I want successful repayments to reduce outstanding balance, so that my loan status stays accurate.
- FR-005 Story KBPS33-18 (5 pts): As a borrower, I want an automated warning only when a repayment is overdue, so that I can act before default.
- NFR-001 Story KBPS33-22 (8 pts): As a platform operator, I want repayment and EMI distribution inside ACID transactions, so that funds cannot double-spend.
- NFR-002 Story KBPS33-26 (8 pts): As a borrower, I want authenticated access and encryption in transit/at rest, so that my financial PII is not disclosed.

## 5. Tasks and sub-tasks under every FR/NFR

Each story has an implementation Task and a verification Sub-task (for example, FR-001 Task: heuristic credit-risk engine; FR-001 Sub-task: unit-test DTI/history/employment rules). Tasks and sub-tasks appear on the same Kanban board under their parent epic.

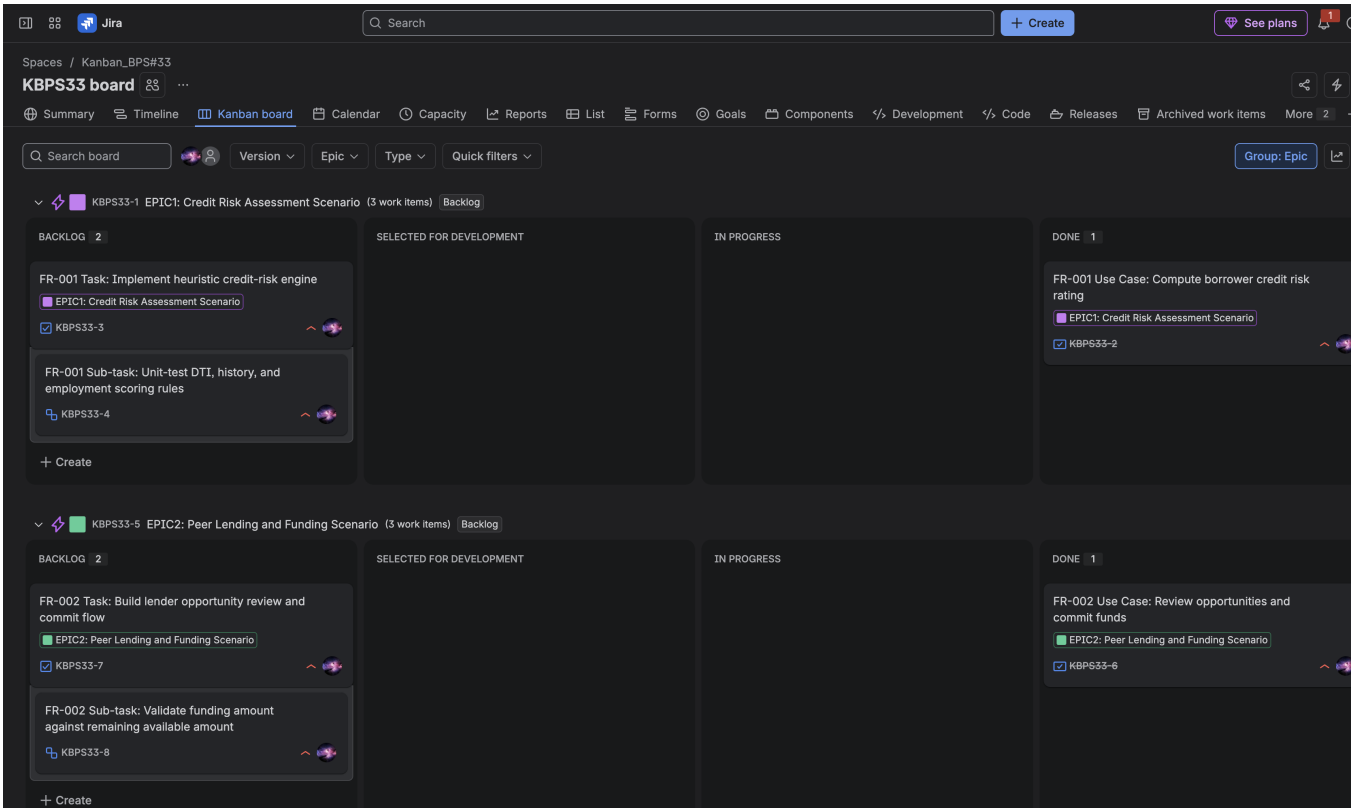



**Figure.** Kanban board continued: additional epic swimlanes with FR tasks/sub-tasks in Backlog and In Progress, and completed use-case stories in Done.

## 6. Backlog with every item linked to an Epic

The backlog/list is ordered by epic. Every story, task and sub-task is parented to its scenario epic so the instructor can filter by FR-001…FR-005 and NFR-001/NFR-002. Priority is High for core lending FRs and Highest for NFR-001 (financial integrity).

## Sequence of screenshots

1) Board creation / FR+NFR cards on the Kanban board. 2) Epics assigned to board work. 3) Hierarchical list of stories, tasks and sub-tasks. 4) Remaining epic swimlanes on the board.

# Scrum_Project-Mudit Chaturvedi_PES1UG24CS279-BPS#33

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Problem Statement #33 — Micro-Lending & Peer Credit Risk Assessor

Jira site: mudit300477.atlassian.net

Project: **Scrum_BPS#33** | Key: **SBPS33** (company-managed Scrum)

Template: Software Development — Scrum | Board: SBPS33

## 1. Scrum project (repeat of Kanban structure)

The same Lab 1 requirements were recreated in a Scrum template project named Scrum_BPS#33 (key SBPS33). Work types: 7 Epics, 7 Stories (use cases), 7 Tasks and 7 Sub-tasks (SBPS33-1 … SBPS33-28).

## 2. Backlog, epics and Fibonacci story points

Stories were estimated with Planning-Poker style Fibonacci points based on complexity, uncertainty and test effort: FR-001 = 8, FR-002 = 5, FR-003 = 5, FR-004 = 5, FR-005 = 5, NFR-001 = 8, NFR-002 = 8 (total 44 points). Sprint 2 committed 26 of those points (FR-004, FR-005, NFR-001, NFR-002).

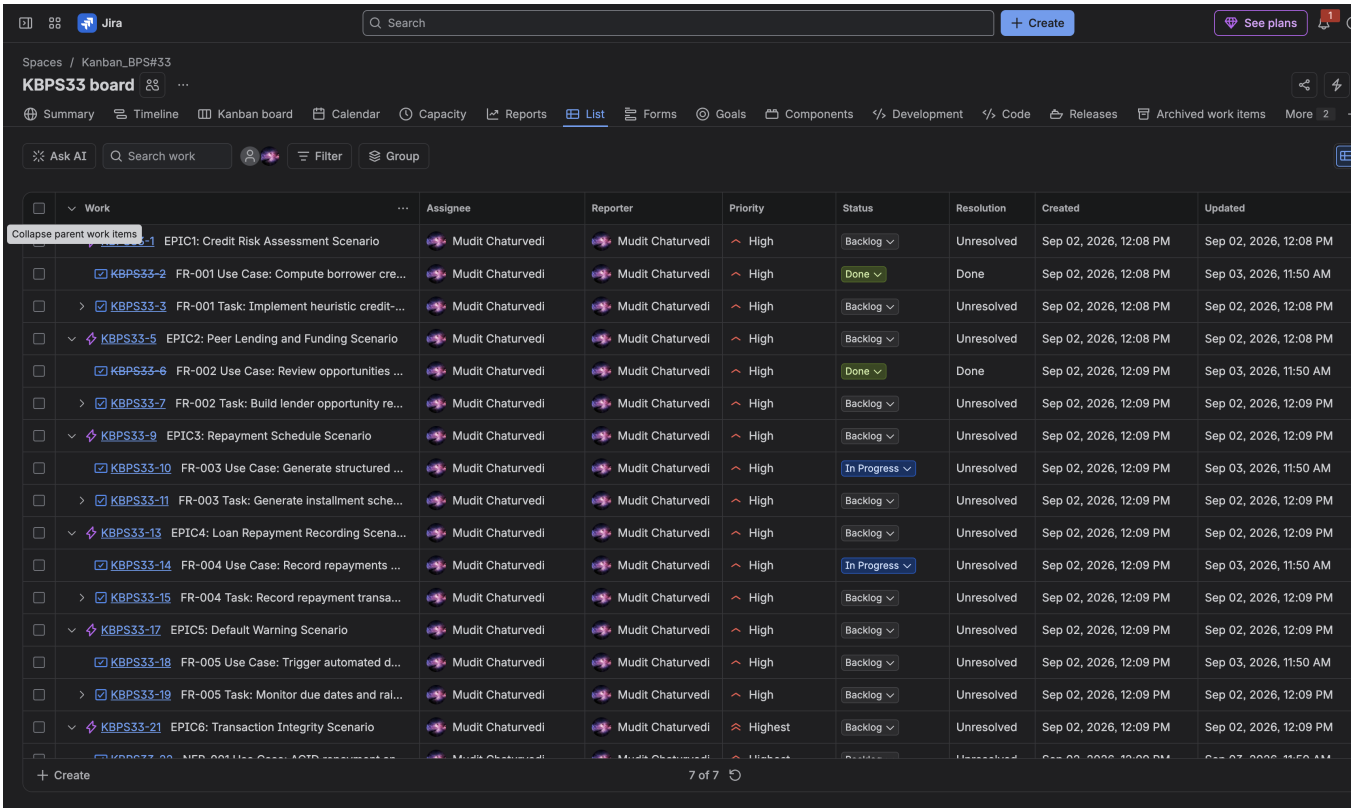



**Figure.** Scrum backlog: Sprint 2 (8 Sep–15 Sep) with epic labels, statuses, and story-point estimates (5 and 8) on FR/NFR use-case stories.

## 3. Sprint simulation (two sprints)

**Sprint 1** (1 week, 1–8 Sep 2026, now closed): goal — deliver FR-001 credit-risk rating, FR-002 peer funding, and FR-003 repayment schedule. Stories SBPS33-2, SBPS33-6 and SBPS33-10 were completed (Done). **Sprint 2** (1 week, 8–15 Sep 2026, active then completed in simulation): goal — repayment recording, default warnings, and NFR integrity/security. Work was moved To Do → In Progress → Done on the Active Sprints board.

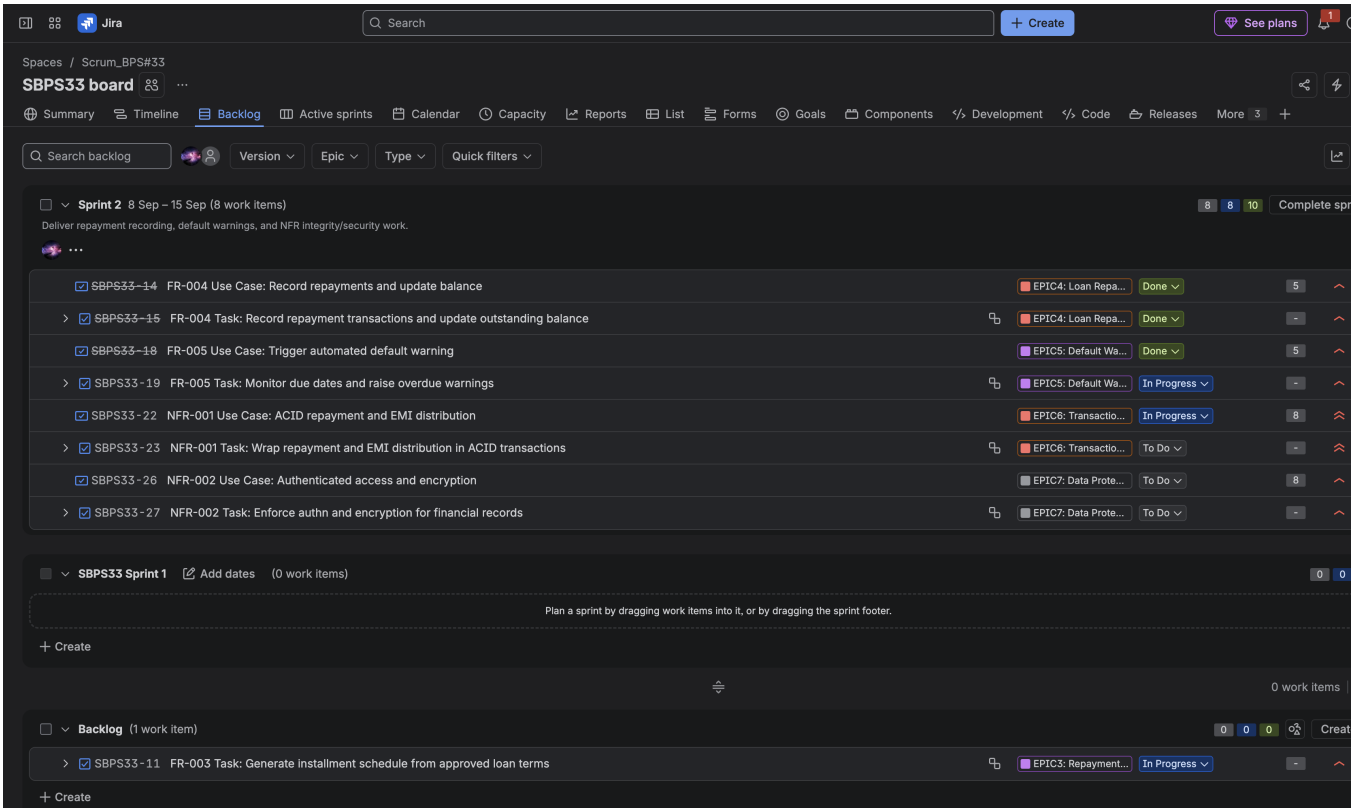



**Figure.** Active Sprint view (Sprint 2): stories and sub-tasks progressed into Done after the simulated sprint workflow.

## 4. Burndown chart

Reports → Burndown Chart, estimated in Story Points. The grey guideline is ideal linear progress; the remaining-values line tracks unfinished points. Sprint 2's chart is flatter than a real team week because story points were applied after sprint start, so the start-of-sprint remaining value logged as 0 and later completions appear as scope/estimate updates. That itself is a useful lab insight: estimates must be on the stories before Start sprint, otherwise burndown cannot show a classic downward slope.

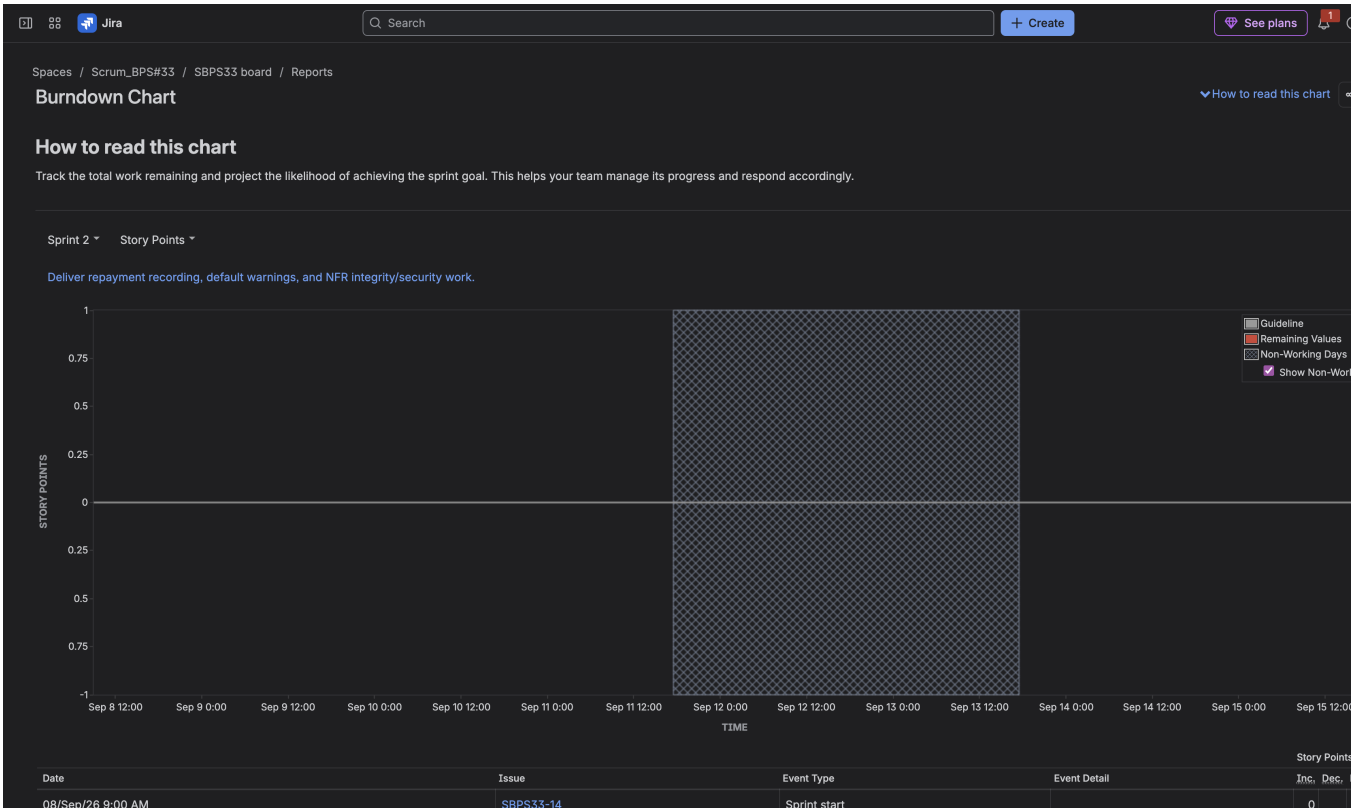



**Figure.** Sprint 2 burndown (Story Points): time axis 8–15 Sep, sprint goal shown, events table listing SBPS33-14 and related items.

## 5. Reflection questions

### 1. Did your estimations reflect the actual effort?

In this simulated sprint the Fibonacci values were relative, not calendar hours. Credit-risk heuristics and ACID/security work (8 points) are genuinely harder than recording a repayment (5). Because the lab compressed a week of work into minutes, the numbers overstate elapsed time but correctly rank uncertainty. Next time we would split any 8-point story before committing it.

### 2. Was your backlog well-prioritized?

Yes. High-value lender/borrower FRs (risk rating, funding, schedule) went into Sprint 1. Recording, warnings, ACID and encryption followed in Sprint 2. That matches user impact: you cannot repay or warn if you cannot score and fund a loan first.

### 3. How did your simulated sprint align with your plan?

Sprint 1 delivered the three planned FR stories. Sprint 2 contained the remaining FR/NFR stories and was moved to Done on the board. One leftover FR-003 task remained in the product backlog, which is realistic incomplete spillover rather than hiding unfinished work inside a closed sprint.

### 4. What insights did the burndown chart give about your team's capacity?

A flat remaining-values line at zero at sprint start means the team did not have a trusted baseline velocity. If a real team always finishes early, they under-committed; if the red line stays above the guideline, they over-committed. For this lab, the chart mainly showed that capacity cannot be read until story points exist at sprint start. That is the operational takeaway for the next sprint planning session.

## 6. Epics summary (handout deliverable)

EPIC1 Credit Risk Assessment; EPIC2 Peer Lending and Funding; EPIC3 Repayment Schedule; EPIC4 Loan Repayment Recording; EPIC5 Default Warning; EPIC6 Transaction Integrity; EPIC7 Data Protection. Each

epic’s stories carry the story-point totals listed above.

# BugReport.PDF

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Project: **Bug_BPS#33** | Key: **BBPS33** (company-managed bug tracker / Kanban)

Eight defects mapped to FR-001…FR-005 and NFR-001/NFR-002

## 1. Bug tracker project

A dedicated company-managed software project Bug_BPS#33 (key BBPS33) was created to report defects against the Micro-Lending platform. The BBPS33 board uses Backlog / Selected for Development / In Progress / Done, with Expedite vs Everything Else swimlanes for severity.

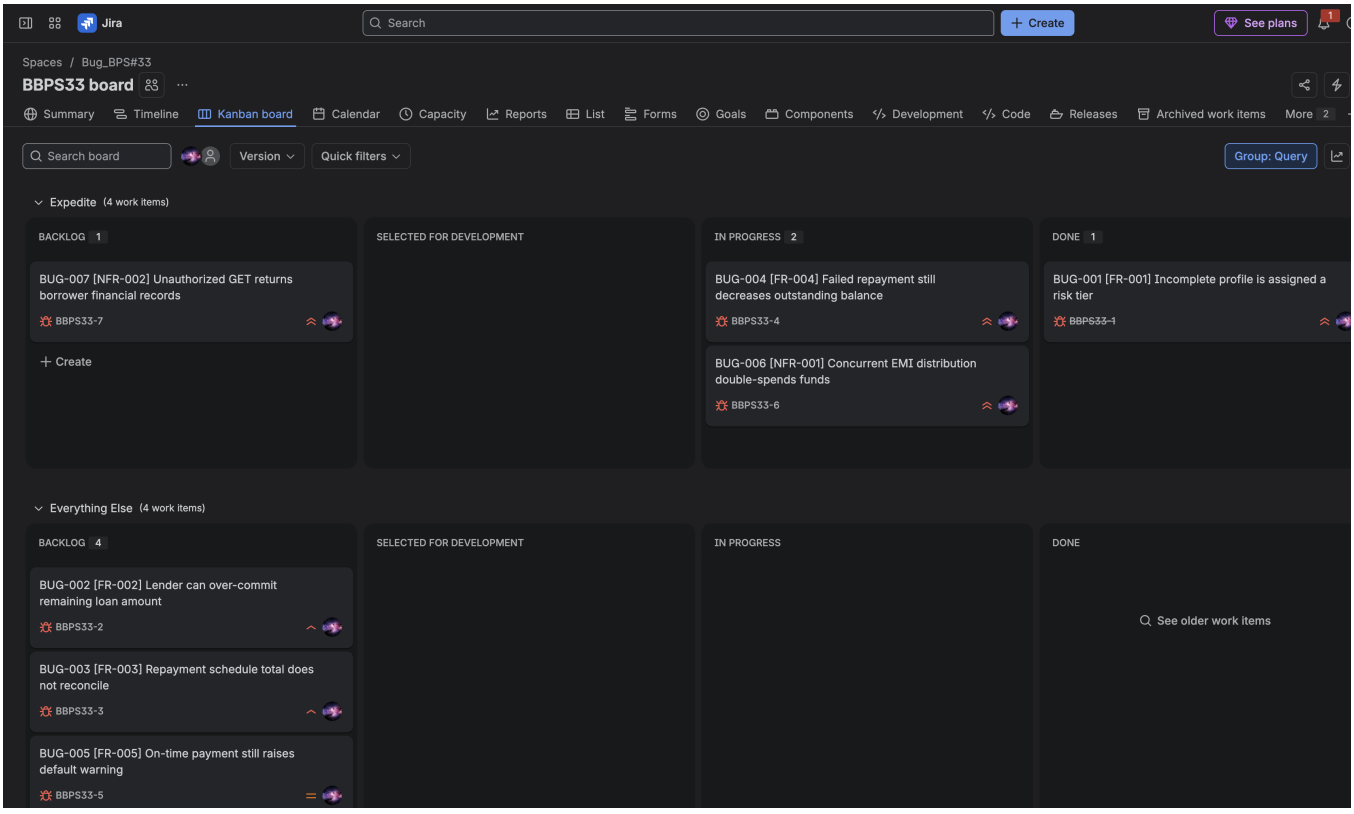



**Figure.** BBPS33 board after all bugs were reported: Expedite lane holds critical FR-001/FR-004/NFR bugs; remaining FR/NFR bugs sit in Backlog.

## 2. Bugs reported (with description)

1 **BBPS33-1 BUG-001 [FR-001] Incomplete profile is assigned a risk tier** — Fail of FR-001 acceptance: incomplete financial profiles must not be rated. Status: Done (fix verified).

2. **BBPS33-2 BUG-002 [FR-002] Lender can over-commit remaining loan amount** — Funding amount is not capped at remaining available amount.
3. **BBPS33-3 BUG-003 [FR-003] Repayment schedule total does not reconcile** — Installment sum does not match principal + interest from loan terms.
4. **BBPS33-4 BUG-004 [FR-004] Failed repayment still decreases outstanding balance** — Failed payment mutates balance; should be no-op. In Progress.
5. **BBPS33-5 BUG-005 [FR-005] On-time payment still raises default warning** — Warning generated for paid-on-time installment.
6. **BBPS33-6 BUG-006 [NFR-001] Concurrent EMI distribution double-spends funds** — Missing ACID boundary; duplicate distribution under concurrency. In Progress.
7. **BBPS33-7 BUG-007 [NFR-002] Unauthorized GET returns borrower financial records** — Authn/authorization bypass on financial API.
8. **BBPS33-8 BUG-008 [NFR-002] Risk API error payload leaks PII in plaintext** — Error responses include unencrypted borrower identifiers.

## 3. How this supports Lab 2

Each bug is the explicit fail path of a Lab 1 acceptance criterion, so the instructor can trace FR/NFR IDs from the requirements table through Scrum/Kanban stories to defect reports.